Explanation, Features and Development of Student Academic Dashboard

Student Name: Ekansh (Roll No: 26001039) | **Branch:** B.Tech – CSM (1st Semester) | **College:** DR. RVR NRI Institute of Technology

1. Introduction

The given code is used to develop a **Student Academic Dashboard** (`03-student.html`) for **Ekansh** (Roll Number: **26001039**), a first-year B.Tech student specializing in Computer Science and Machine Learning (**CSM**) at **DR. RVR NRI Institute of Technology, Vijayawada, Andhra Pradesh**, for the Academic Year 2026–2027.

The website is created using **HTML5** and **CSS3**. It presents a comprehensive student portal featuring a welcome greeting, 4 numerical metric overview cards, a 6-item student profile grid, 5 subject performance cards, structured tables for internal marks, attendance reports, and assignments with color-coded status badges, a complete class timetable, and academic notices.

The dashboard uses a clean professional gray and light-blue theme (`#f5f6f8` background, `#ffffff` panels, `#e8eef7` welcome banners, and `#334e73` navy accents). It is built with CSS Grid (`grid-template-columns: repeat(auto-fit, minmax(...))`) and responsive wrappers to ensure seamless rendering across desktop monitors, tablets, and mobile smartphones.

2. Explanation of the Code

The dashboard is developed using two main front-end web technologies: **HTML** and **CSS**.

A. HTML (HyperText Markup Language)

HTML is used to structure the academic dashboard elements, navigation jump links, metric overview cards, profile items, subject grids, data tables, and notices. The main HTML elements used in `03-student.html` are:

HTML Element	Purpose in Ekansh's Student Academic Dashboard
`<!DOCTYPE html>`	Declares the document as an HTML5 webpage.
`<html lang="en">`	Defines the root element and English language setting of the document.
`<head>, <meta>`	Contains charset settings, viewport responsive scaling, and the internal `<style>` block.
`<title>`	Sets the browser tab title to *"Student Academic Dashboard"*.
`<body>`	Contains all visible dashboard sections and footer credits.
`<header>`	Displays the main portal title (`Student Academic Dashboard`) and institutional affiliation (`DR. RVR NRI Institute of Technology`).
`<nav>, <a>`	Provides the top navigation bar with anchor links to `Dashboard`, `Profile`, `Subjects`, `Marks`, `Attendance`, `Assignments`, `Timetable`, and `Notices`.
`<main class="container">`	Centers and constrains all dashboard content within a maximum width of `1200px`.
`<section>`	Organizes the dashboard into distinct semantic modules for each academic report.
`<div>`	Groups elements for card grids, welcome banners, profile grids, and table wrappers.
`<h1>, <h2>, <h3>`	Displays portal headers, section titles (`.section-title`), and card headings.
`<table>, <tr>, <th>, <td>`	Creates structured data tables for Marks Report, Attendance Report, Assignments Tracker, and Class Timetable.
`<span class="status">`	Displays color-coded assignment status badges (`submitted`, `progress`, `pending`).
`<footer>`	Displays student credentials, roll number, branch, semester, and academic year at the bottom of the portal.

B. CSS (Cascading Style Sheets)

CSS is used to control the academic color scheme, typography, CSS Grid card layouts, table styling, and responsive mobile adaptation. The important CSS properties used in `03-student.html` are:

CSS Property	Purpose in Ekansh's Student Academic Dashboard
`* { box-sizing: border-box; }`	Ensures padding and borders are included in element width calculations across all dashboard boxes.
`background-color`	Sets dashboard background (`#f5f6f8`), panel surfaces (`#ffffff`), welcome banners (`#e8eef7`), and profile items (`#f5f7fa`).
`color`	Defines text colors for main text (`#333333`), headings (`#334e73`), card subtitles (`#606b78`), and numeric values.
`display: grid`	Arranges overview cards, profile items, and subject boxes using `grid-template-columns: repeat(auto-fit, minmax(...))`.
`overflow-x: auto`	Wraps wide data tables in `.table-wrap` to enable horizontal scrolling on small mobile screens without breaking page layout.
`border-collapse: collapse`	Combines table cell borders into clean single lines across all four dashboard tables.
`tbody tr:nth-child(even)`	Applies alternating row background shading (`#f8f9fb`) to improve table readability.
`Status Badge Colors`	Styles assignment statuses with soft background fills: Green (`#e2f3e8`), Yellow (`#fff1d6`), and Red (`#fbe5e5`).
`border-left: 3px solid`	Adds left accent indicator bars to academic notice boxes in the Notices & Reminders module.
`border-radius`	Rounds corners of cards, panels, profile boxes, and status badges (`4px` to `8px`).
`:hover pseudo-class`	Underlines navigation links and shifts text color to dark navy (`#334e73`) on hover.
`@media (max-width: 600px)`	Adjusts header padding, navigation gap spacing, container widths (92%), and section title sizes for mobile viewports.

3. Features of the Website

1. Header, Navigation Bar & Welcome Module (#dashboard)

Displays the top portal title (`Student Academic Dashboard — DR. RVR NRI Institute of Technology`), an 8-link navigation bar, a personalized welcome greeting (*"Welcome, Ekansh! — Branch: CSM | Semester: 1st Semester"*), and four numerical metric cards:
Total Subjects
(5),
Average Attendance
(`90.2%`),
Average Marks
(`43/50`), and
Assignments Submitted
(`3/5`).

2. Student Profile Section (#profile)

Presents a structured 6-item profile grid displaying verified student details: Name (
Ekansh
), Roll Number (
26001039
), College (
DR. RVR NRI Institute of Technology
), Branch (
CSM
), Semester (

4. Development of the Website

The Student Academic Dashboard is developed through the following systematic steps:

Step 1: Create the HTML file — Create a file named `03-student.html`. Define the HTML5 document structure and build the header, navigation bar, welcome banner, 4 overview metric cards, profile grid, subject cards, 4 data tables (Marks, Attendance, Assignments, Timetable), notices, and footer.

Step 2: Add CSS styling — Inside the `<head>` section, add a `<style>` block setting the neutral gray background (`#f5f6f8`), white card surfaces (`#ffffff`), navy heading colors (`#334e73`), border styles, and status badge background colors.

Step 3: Design CSS Grid layouts — Use CSS Grid (`display: grid; grid-template-columns: repeat(auto-fit, minmax(...));`) for the overview cards (`.cards`), profile grid (`.profile-grid`), and subject cards (`.subject-grid`).

Step 4: Build tables and responsive wrappers — Create structured HTML tables for marks, attendance, assignments, and timetable, and wrap each table inside `.table-wrap` with `overflow-x: auto` to support mobile devices.

Step 5: Make the dashboard responsive — Add a `@media (max-width: 600px)` query to adjust header padding, navigation gap spacing, container widths (92%), and panel padding for smartphones.

Step 6: Test the website — Open `03-student.html` in a web browser. Verify section jump navigation, table row shading, status badge colors, and mobile screen responsiveness.

5. Website Folder Structure

Keep the project files organized in your local directory and GitHub repository as shown below:

```
Ekansh-Student-Dashboard/ ├── 03-student.html # Student Academic Dashboard webpage and internal CSS ├── 01-portfolio.html # Project 01: Ekansh Personal Portfolio Website ├── 02-college.html # Project 02: DR. RVR NRI College Website & Admissions ├── README.md # GitHub repository documentation and project guide └── Student_Dashboard_Documentation.pdf # Complete explanation, features, output, and code PDF
```

6. Technologies Used

Technology	Usage
HTML5	Creates the academic modules, profile grid, data tables, and notices.
CSS3	Controls the professional gray and navy color palette, typography, borders, and spacing.
CSS Grid	Arranges overview metric cards, profile items, and subject cards into responsive columns.
HTML Tables & Badges	Structures marks, attendance, assignments, and timetable data with color-coded status badges.
Responsive Media Queries	Adapts header padding, container widths, and font sizes for mobile viewports.
Web Browser	Renders and runs the static academic dashboard directly on any device.

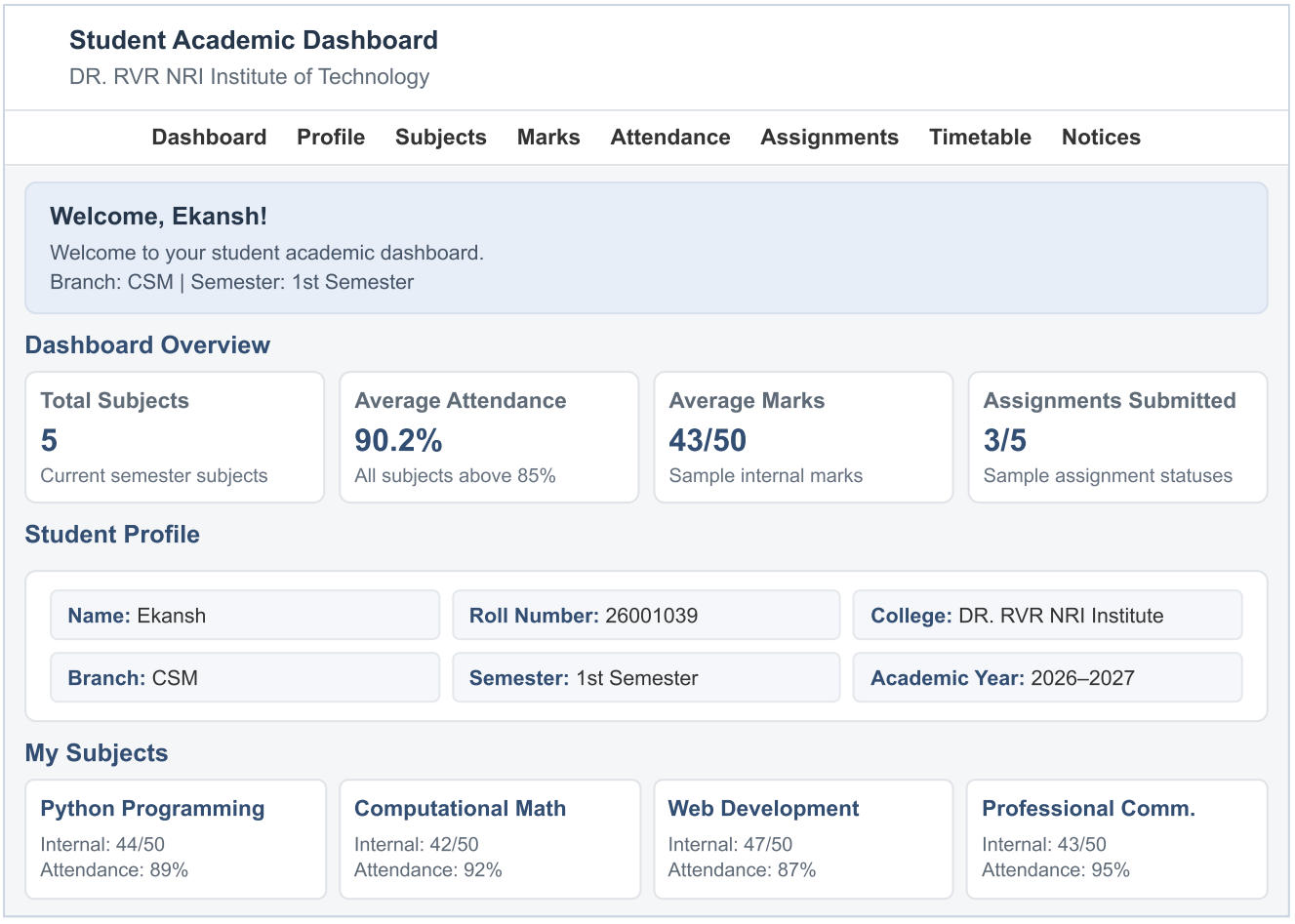
The dashboard is a static front-end web application. It uses CSS Grid and table wrappers for layout flexibility and runs instantly in any modern web browser without requiring a backend database.

7. Advantages of the Website

1. Provides Ekansh with a centralized portal to view all semester academic data.
2. Displays key academic statistics (attendance, marks, assignments) instantly via overview cards.
3. Organizes student profile details (Roll No: 26001039, CSM branch) in a clean grid format.
4. Tracks internal marks and attendance percentages across all 5 semester subjects.
5. Uses color-coded status badges for clear assignment deadline tracking.
6. Includes a detailed daily class timetable from 9:20 AM to 4:25 PM.

9. Output (Part 1 — Welcome, Overview Cards, Profile & Subjects)

Below is the rendered browser output of the top sections of `03-student.html`:



9. Output (Part 2 — Marks, Attendance, Assignments, Timetable & Footer)

Below is the rendered browser output of the tables and notices in `03-student.html`:

Marks Report

S.No	Subject	Obtained	Max	Performance
1	Python Programming	44	50	Good
2	Computational Mathematics	42	50	Good
3	Web Development	47	50	Excellent

Assignments Status

S.No	Subject	Assignment	Status
1	Python Programming	List Methods and Programs	Submitted
2	Computational Math	Problem Solving Worksheet	In Progress
3	Web Development	HTML and CSS Website	Submitted

Notices & Reminders

Assignment Reminder
Check assignment deadlines and complete pending work on time.

Attendance Reminder
Maintain good attendance in every subject throughout the semester.

Student Academic Dashboard
Ekansh | Roll No: 26001039 | CSM | 1st Semester | Academic Year 2026–2027

10. Code (Part 1 — HTML Head & CSS Styling)

Complete source code of `03-student.html` (Student Academic Dashboard):

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-
width, initial-scale=1.0">
<title>Student Academic Dashboard</title>

<style>
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
    font-family: Arial, sans-serif;
}

html {
    scroll-behavior: auto;
}

body {
    background: #f5f6f8;
    color: #333333;
    line-height: 1.6;
}

header {
    background: #ffffff;
    color: #26364a;
    padding: 25px 7%;
    border-bottom: 1px solid #e0e3e8;
}

header h1 {
    margin-bottom: 5px;
}

nav {
    background: #ffffff;
    padding: 14px 7%;
    display: flex;
    flex-wrap: wrap;
    gap: 20px;
    border-bottom: 1px solid #e0e3e8;
}

nav a {
    color: #333333;
    text-decoration: none;
    font-weight: bold;
    font-size: 14px;
}

nav a:hover {
    color: #334e73;
    text-decoration: underline;
}

.container {
    width: 86%;
    max-width: 1200px;
    margin: 30px auto;
}

.welcome {
    background: #e8eef7;
    color: #26364a;
    padding: 28px;
    border: 1px solid #d7e0ed;
    border-radius: 8px;
    margin-bottom: 25px;
}

.welcome h2 {
    margin-bottom: 8px;
}

.section-title {
    color: #334e73;
    margin: 32px 0 16px;
    font-size: 23px;
}

.cards {
    display: grid;
    grid-template-columns: repeat(auto-fit,
minmax(200px, 1fr));
    gap: 18px;
}

.card {
    background: #ffffff;
    padding: 20px;
    border-radius: 8px;
    border: 1px solid #e0e3e8;
}

.card h3 {
    color: #606b78;
    font-size: 15px;
    margin-bottom: 8px;
}

.number {
    color: #334e73;
    font-size: 27px;
    font-weight: bold;
}

.card p {
    margin-top: 5px;
    color: #687382;
    font-size: 14px;
}

.panel {
    background: #ffffff;
    padding: 22px;
    margin-bottom: 22px;
    border: 1px solid #e0e3e8;
    border-radius: 8px;
}
```

10. Code (Part 2 — Remaining CSS & Dashboard Components)

```
.profile-grid {
    display: grid;
    grid-template-columns: repeat(auto-fit,
minmax(230px, 1fr));
    gap: 14px;
}

.profile-item {
    padding: 12px;
    background: #f5f7fa;
    border: 1px solid #e5e8ed;
    border-radius: 5px;
    overflow-wrap: anywhere;
}

.profile-item strong {
    color: #334e73;
}

.subject-grid {
    display: grid;
    grid-template-columns: repeat(auto-fit,
minmax(200px, 1fr));
    gap: 18px;
}

.subject {
    background: #ffffff;
    padding: 20px;
    border: 1px solid #e0e3e8;
    border-radius: 8px;
}

.subject h3 {
    color: #334e73;
    margin-bottom: 10px;
}

.subject p {
    color: #606b78;
    font-size: 14px;
    margin-top: 5px;
}

.table-wrap {
    width: 100%;
    overflow-x: auto;
}

table {
    width: 100%;
    min-width: 580px;
    border-collapse: collapse;
    background: #ffffff;
}

th, td {
    padding: 13px;
    text-align: left;
    border: 1px solid #e0e3e8;
}

th {
    background: #e8eef7;
    color: #26364a;
}
```

```
.status {
    display: inline-block;
    padding: 5px 9px;
    border-radius: 4px;
    font-size: 12px;
    font-weight: bold;
}

.submitted {
    background: #e2f3e8;
    color: #24643b;
}

.progress {
    background: #fff1d6;
    color: #805900;
}

.pending {
    background: #fbe5e5;
    color: #9b3030;
}

.notice {
    padding: 14px;
    background: #f5f7fa;
    border-left: 3px solid #879bb8;
    margin-bottom: 12px;
}

.notice p {
    margin-top: 5px;
    color: #606b78;
}

footer {
    background: #e8eef7;
    color: #26364a;
    text-align: center;
    padding: 22px;
    margin-top: 35px;
    border-top: 1px solid #d7e0ed;
}

@media (max-width: 600px) {
    header { padding: 22px 5%; }
    nav { padding: 14px 5%; gap: 12px; }
    .container { width: 92%; }
    .welcome { padding: 20px; }
    .panel { padding: 14px; }
    .section-title { font-size: 21px; }
}
</style>
</head>
```

10. Code (Part 3 — Complete HTML Body of 03-student.html)

```
<body>
<header>
    <h1>Student Academic Dashboard</h1>
    <p>DR. RVR NRI Institute of
Technology</p>
</header>

<nav>
    <a href="#dashboard">Dashboard</a>
    <a href="#profile">Profile</a>
    <a href="#subjects">Subjects</a>
    <a href="#marks">Marks</a>
    <a href="#attendance">Attendance</a>
    <a href="#assignments">Assignments</a>
    <a href="#timetable">Timetable</a>
    <a href="#notices">Notices</a>
</nav>

<main class="container">
    <section id="dashboard">
        <div class="welcome">
            <h2>Welcome, Ekansh!</h2>
            <p>Welcome to your student
academic dashboard.</p>
            <p>Branch: CSM | Semester: 1st
Semester</p>
        </div>
        <h2 class="section-title">Dashboard
Overview</h2>
        <div class="cards">
            <div class="card">
                <h3>Total Subjects</h3>
                <div class="number">5</div>
                <p>Current semester
subjects</p>
            </div>
            <div class="card">
                <h3>Average Attendance</h3>
                <div class="number">90.2%
</div>
                <p>All subjects above 85%
</p>
            </div>
            <div class="card">
                <h3>Average Marks</h3>
                <div
class="number">43/50</div>
                <p>Sample internal marks</p>
            </div>
            <div class="card">
                <h3>Assignments
Submitted</h3>
                <div
class="number">3/5</div>
                <p>Sample assignment
statuses</p>
            </div>
        </div>
    </section>

    <section id="profile">
        <h2 class="section-title">Student
Profile</h2>
        <div class="panel">
            <div class="profile-grid">
                <div class="profile-item">
    <section id="subjects">
        <h2 class="section-title">My
Subjects</h2>
        <div class="subject-grid">
            <div class="subject"><h3>Python
Programming</h3><p>Internal Marks: 44/50</p>
<p>Attendance: 89%</p></div>
            <div class="subject">
<h3>Computational Mathematics</h3>
<p>Internal Marks: 42/50</p><p>Attendance:
92%</p></div>
            <div class="subject"><h3>Web
Development</h3><p>Internal Marks: 47/50</p>
<p>Attendance: 87%</p></div>
            <div class="subject">
<h3>Professional Communication</h3>
<p>Internal Marks: 43/50</p><p>Attendance:
95%</p></div>
            <div class="subject">
<h3>CTLR</h3><p>Internal Marks: 39/50</p>
<p>Attendance: 88%</p></div>
        </div>
    </section>

    <section id="marks">
        <h2 class="section-title">Marks
Report</h2>
        <div class="panel"><div
class="table-wrap"><table>
            <thead><tr><th>S.No.</th>
<th>Subject</th><th>Marks Obtained</th>
<th>Maximum Marks</th><th>Performance</th>
</tr></thead>
            <tbody>
                <tr><td>1</td><td>Python
Programming</td><td>44</td><td>50</td>
<td>Good</td></tr>
                <tr><td>2</td>
<td>Computational Mathematics</td>
<td>42</td><td>50</td><td>Good</td></tr>
                <tr><td>3</td><td>Web
Development</td><td>47</td><td>50</td>
<td>Excellent</td></tr>
                <tr><td>4</td>
<td>Professional Communication</td>
<td>43</td><td>50</td><td>Good</td></tr>
                <tr><td>5</td><td>CTLR</td>
<td>39</td><td>50</td><td>Good</td></tr>
            </tbody>
        </table></div></div>
    </section>

    <section id="attendance">
        <h2 class="section-title">Attendance
Report</h2>
        <div class="panel"><div
class="table-wrap"><table>
            <thead><tr><th>S.No.</th>
<th>Subject</th><th>Classes Attended</th>
<th>Total Classes</th><th>Attendance</th>
</tr></thead>
            <tbody>
                <tr><td>1</td><td>Python
Programming</td><td>40</td><td>45</td>
<td>89%</td></tr>
                <tr><td>2</td>
```